Data 604

Working with Data at Scale

Analyzing the Impact of Oil Prices on Canadian Energy Stocks

Team Name: Team 2

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Introduction

Alberta's economny is substantially dependent on its Oil and Energy sector.The motivation behind this project is to explore the relationship between stock prices of these oil companies and various macroeconomic indicators to gain insights into how external economic factors influence market movements. Stock prices are affected by a multitude of factors, including oil prices, inflation rates, GDP growth, and geopolitical events. Understanding these relationships is crucial for investors, policymakers, and financial analysts who seek to interpret market trends. Our key research questions include:

- How do macroeconomic indicators correlate with stock price movements?
- Which macroeconomic factors have the strongest influence on stock volatility?
- How do different companies in the energy sector react to external economic changes?

We chose this project because of its relevance to financial decision-making and risk assessment in the oil and gas industry, which plays a crucial role in the Canadian economy.

Note##

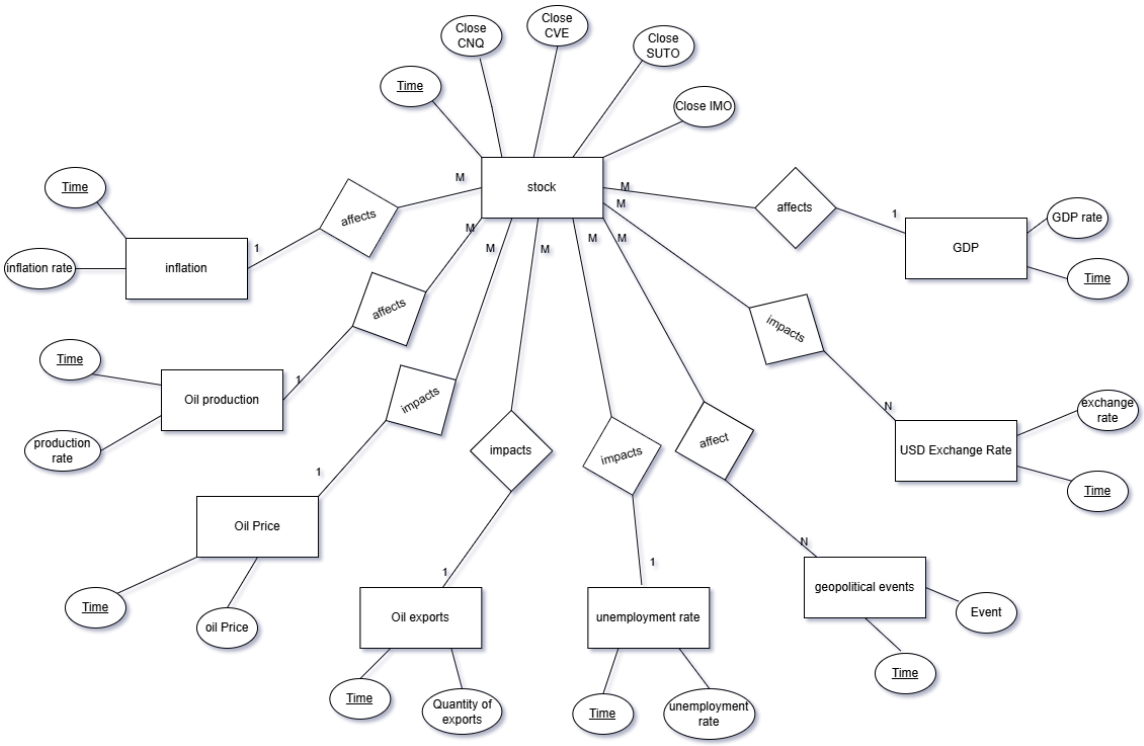
- We have attached our work in a Jupyter Notebook titled "Final_Project_Report-604.ipynb".
- All references to query names within the submitted notebook point to "Final_Project_Report-604.ipynb".
- Additionally, we have included the dataset files to facilitate running the Jupyter Notebook if needed.

Datasets

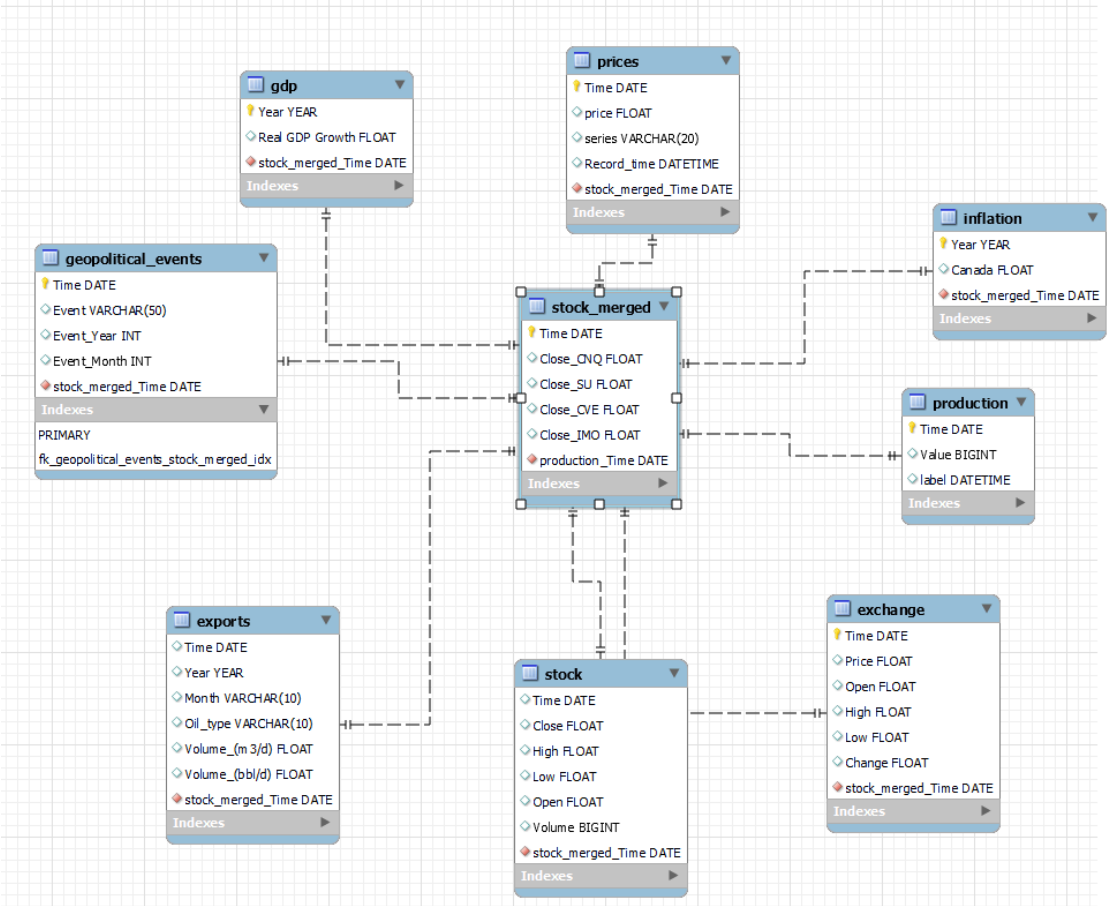
Each team member was responsible for analyzing two macroeconomic indicators. We combined these datasets with a common stock price table, allowing us to analyze correlations and trends effectively. Below are the datasets we worked with:

1. Stock Price Data (Common dataset)
    - Contains daily closing prices of major Canadian energy companies.
    - Used as the primary reference for analysis.
2. Macroeconomic Indicators (Assigned to individual members)
    - Oil Prices: Daily crude oil price data.
    - Inflation Rate: Monthly inflation rates affecting purchasing power.
    - Unemployment Rate: A measure of economic health.
    - GDP Growth: Quarterly GDP trends to analyze economic expansion or recession.
    - Geopolitical Events: Major global events impacting the stock market.

Entity-Relationship Diagram



Database Schema



Data Exploration

The stock data for four major oil companies—Cenovus Energy Inc., Suncor Energy Inc., Canadian Natural Resources Ltd., and Imperial Oil Ltd.—were consistently used across the analysis. However, the macroeconomic indicators varied and divided among team. All analyses were conducted using the closing prices of these stocks.

Distribution of Macroecomic Indicators

- GDP and Unemployment Rate - Kajal
- Geopolitical events and Oil Production - Romith
- Oil Prices and USA-CAD exchange rates - Ali
- Inflation rate and Exports - Hawa

Initial Approach

We started by establishing a connection to a MySQL database using `mysql.connector` to ensure access to the required data. We then imported stock data for companies from CSV files and checked for any missing values. Next, we retrieved relevant stock data from the MySQL database and stored it in Pandas DataFrames for further analysis. To prepare the

data, we identified and handled missing or null values and performed necessary transformations to ensure it was ready for analysis.

Queries and Analysis Results

- Created tables for multiple stock datasets
- Loaded data from CSV files into MySQL.
- 4 tables were created namely:

stock_SU stock_CVE stock_IMO stock_CNQ *(Refer to the submitted notebook for query)*

Merged all the stock tables into one table, named `stock_merged` (Refer to the submitted notebook)

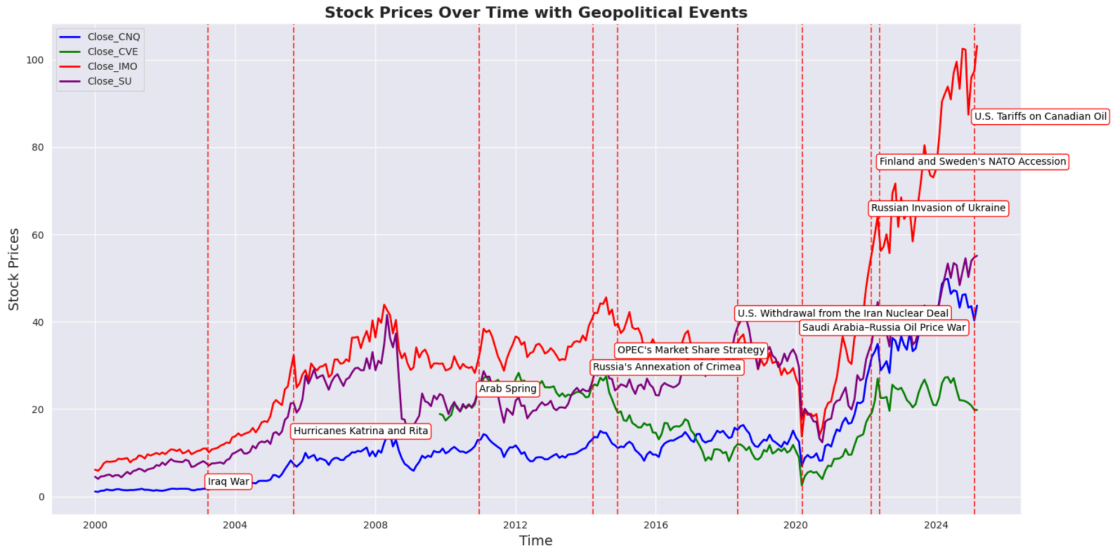
- Each individual created the table for assigned macroeconomic indicator, and used for further analysis by inserting the data into one table.

Guiding questions :

Impact of Geopolitical events on stocks over time:

(Refer to `events_query` from submitted jupyter notebook)

Result :



Findings:

- The plot highlights the impact of geopolitical events on stock prices over time.

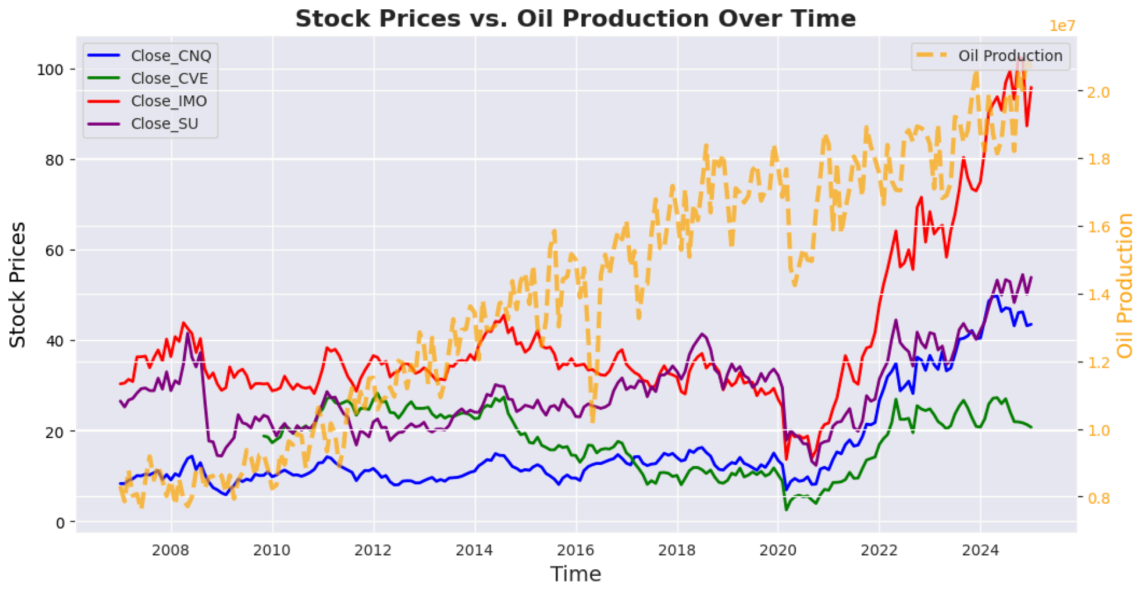
- Major events such as the Iraq War, the Arab Spring, and Russia's annexation of Crimea coincide with fluctuations in stock prices.
- Notably, oil-related events like OPEC's market strategy and teh Saudi arabia- Russia oil price war lead to sharp declines.
- The Russion invasion of Ukraine and Finland & Sweden's NATO accession align with significant stock price surges, particularly for IMO
- This suggests that geopolitical instability and oil market disruptions strongly influence stock price trends.

Stock Prices vs. Oil Production Over Time

(Refer to `production_query` *from submitted jupyter notebook)* Result:

:

	Time	Close_CNQ	Close_CVE	Close_IMO	Close_SU	Value
0	2007-01-01	8.35893	NaN	30.3934	26.5917	8312277
1	2007-02-01	8.34331	NaN	30.5691	25.2926	7865487
2	2007-03-01	9.05646	NaN	31.3971	26.7899	8785947
3	2007-04-01	9.40933	NaN	30.9203	27.1821	8030452
4	2007-05-01	10.11350	NaN	36.3487	28.3542	8075145



Findings:

- The plot shows a strong positive correlation between oil production (oranges dashed line) and the stock prices of CNQ, CVE, IMO, and SU over time.
- Stock prices exhibit significant growth after 2020, particularly for IMO and CNQ, aligning with rising oil production.

- However, stock prices display volatility , with noticeable dips around 2008-2009 and 2020 likely reflectingeconomic downturns or oil market shocks.

Which company stocks are most volatile?

Suggestion made on Discussion portal

(Refer to the `volatility_query` in our submitted notebook) We ranked the companies based on their volatility. Result:

	company	volatility	Volatility_rank
0	CVE	12.547063	1
1	CNQ	9.536057	2
2	SUTO	8.540649	3
3	IMO	7.919516	4

Ranking the average closing prices of oil companies

(Refer to the `close_rank_query` in our submitted notebook) Results:

	company	Avg_close	Close_rank
0	IMO	33.621745	1
1	SUTO	24.101774	2
2	CVE	17.999128	3
3	CNQ	12.857616	4

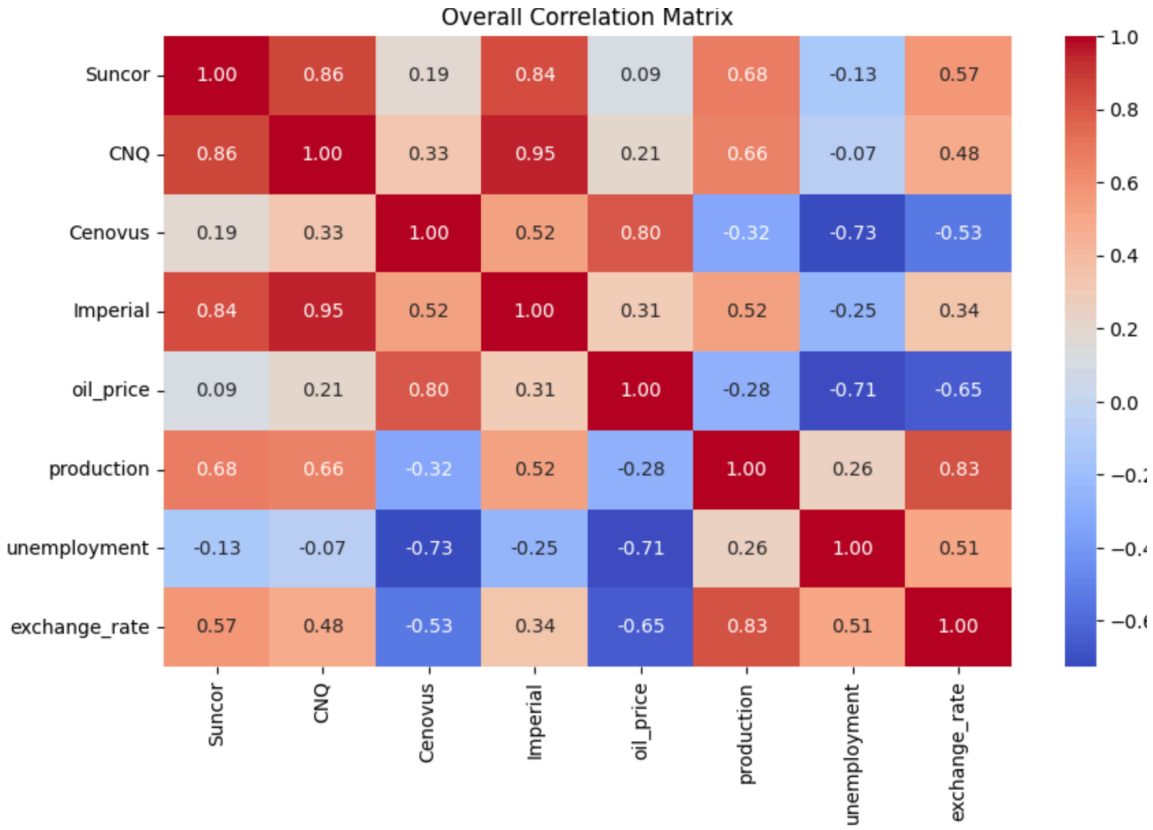
Overall Correlation Matrix

*** Suggestion made on discussion portal***

We have created an overall Corelation matrix for all the macroeconomic indicators.

(Refer to `correlation_query_1` in our submitted notebook)

Results:



Findings:

Highest Correlations (Strongest Positive Relationships):

- CNQ & Imperial (0.95) - These two stocks are highly correlated

Suncor & CNQ (0.86) - Strong correlation

- Oil Price & Cenovus (0.80) - Oil price fluctuations have a strong impact on Cenovus stock.
- Production & Exchange Rate (0.83) - production levels and exchange rates are strongly related.
- Suncor & Imperial (0.84) - Suggests these companies' stock prices move in sync.

Lowest Correlations (Strongest Negative Relationships):

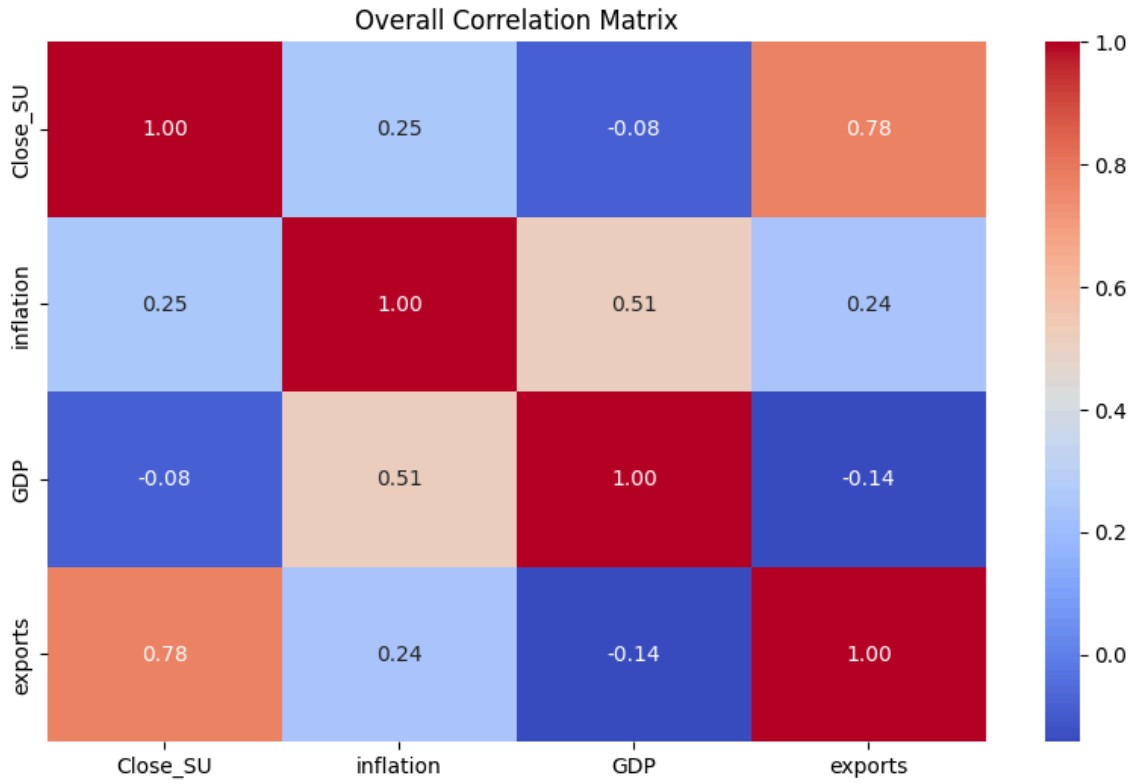
- Cenovus & Unemployment (-0.73) - When unemployment rises, Cenovus stock tends to drop significantly.

- Oil Price & Unemployment (-0.71) - Higher oil prices are associated with lower unemployment.
- Oil Price & Exchange Rate (-0.65) -as oil prices rise, the exchange rate tends to decrease.
- Cenovus & Exchange Rate (-0.53) - Indicates a negative relationship between Cenovus stock and exchange rates.
- Imperial & Unemployment (-0.25) - Shows that Imperial stock is weakly negatively correlated with unemployment.

Relationships between stock prices, oil exports, inflation, and GDP growth over time

(Refer to `correlation_query_2` in our submitted notebook)

Result:



Findings:

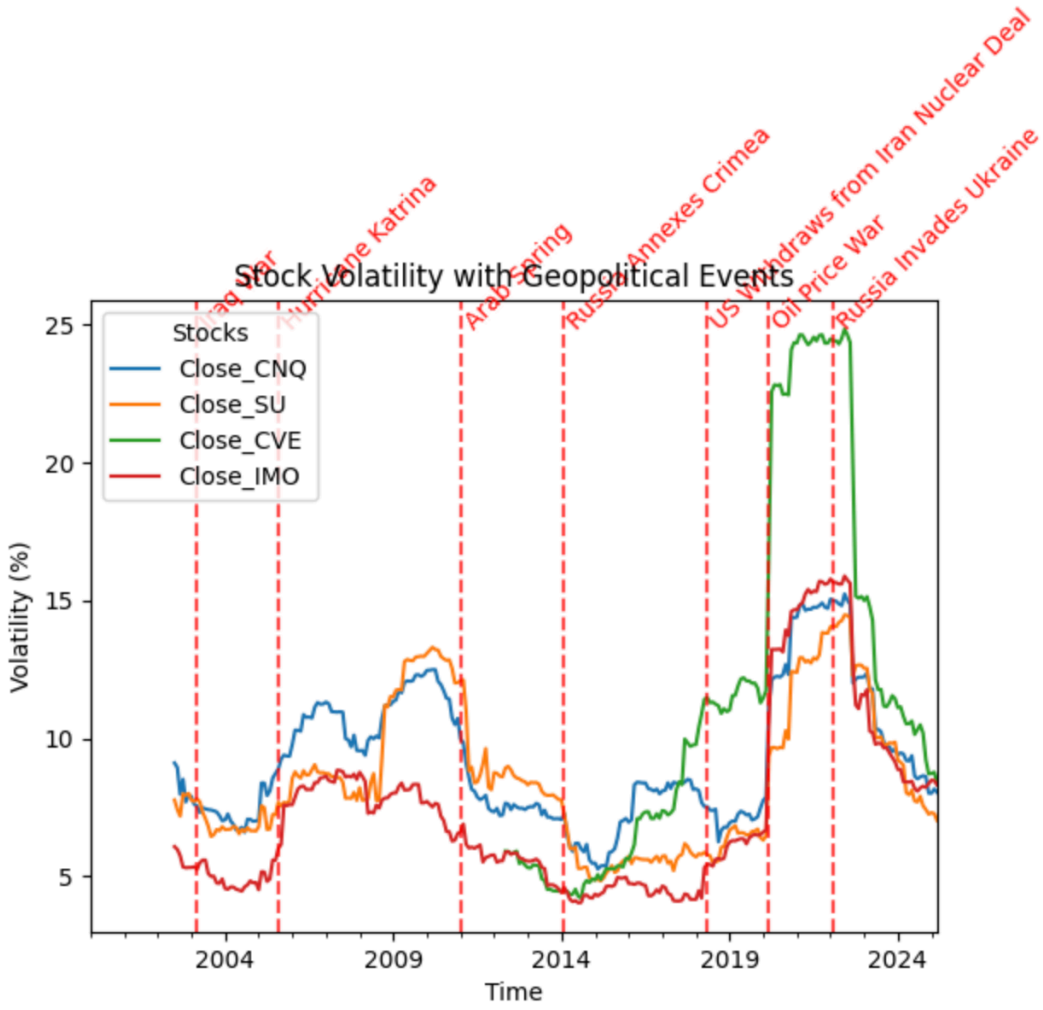
- Close_SU and exports (0.78) indicate a strong positive relationship
- GDP and Close_SU (-0.08) show almost no relationship
- Inflation and GDP (0.51) suggest a moderate relationship
- GDP and exports (-0.14) indicate a weak negative association

Stock Volatility with Geopolitical events

__Suggestion made on discussion portal__

As mentioned earlier , we have created a table stock_merged consisting of al the stock data of all companiee, converted the table to csv and plotted stock Volatility with Geopolitical events. We Computed the daily percentage change in stock prices as a measure of volatility, applied a 30-day rolling standard deviation to smooth fluctuations over time.

Results :



Findings:

- Highest Impact – The Russia-Ukraine invasion of 2022 resulted in a maximum increase in volatility, as opposed to CVE that increased by more than 25%.
- US withdrawal from Iran Nuclear Deal (2018) and Oil Price War (2020) also resulted in considerable stock volatility, specifically impacts CVE and CNQ.
- Lowest Impact – Hurricane Katrina (2005) and Arab Spring (2010-2012) registered relatively lower volatility impacts on all stocks.
- General Trend – Geopolitical happenings related to oil generally exert a high influence on volatility, and CVE is always the most volatile amongst all stocks.

Which is Most Impactful Geopolitical Events on Stock Volatility?

(Refer to `event_impact` from our submitted notebook)

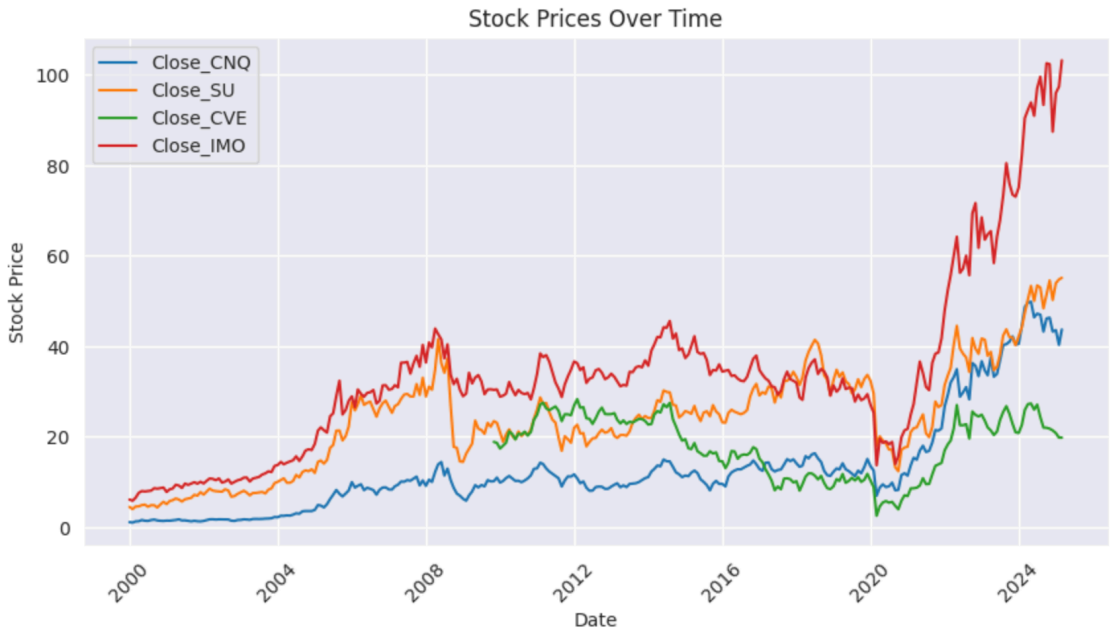
Result:

```
Most Impactful Geopolitical Events on Stock Volatility:
Russia Invades Ukraine: 24.42%
Oil Price War: 16.72%
Arab Spring: 12.12%
US Withdraws from Iran Nuclear Deal: 11.48%
Hurricane Katrina: 9.25%
Iraq War: 7.70%
Russia Annexes Crimea: 6.69%
```

Stock Prices Over time

Plotted an overall trend for stock prices.

Results:



Findings:

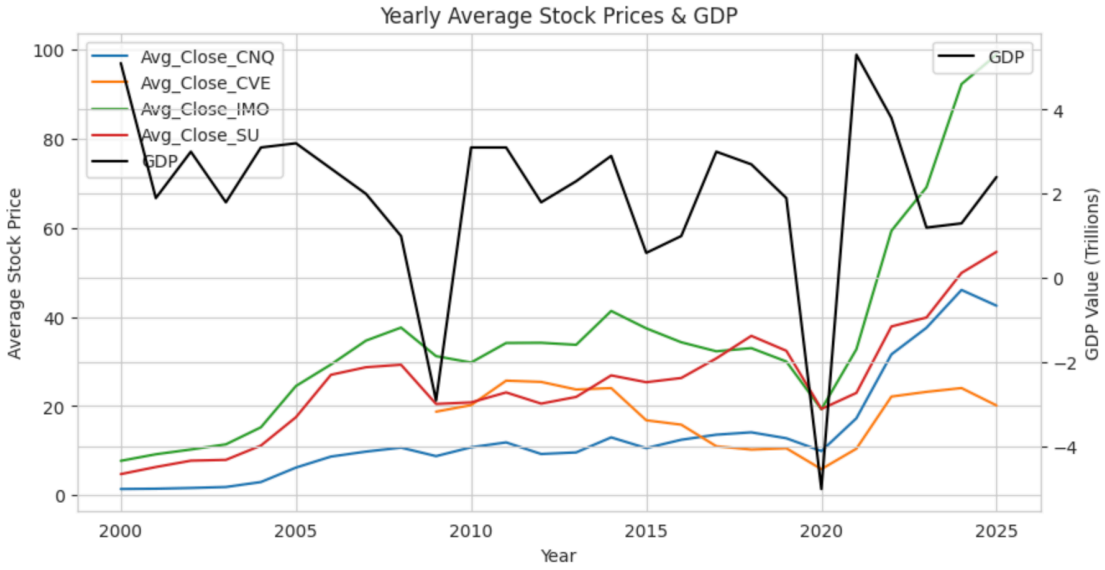
- The stock prices of CNQ, SU, CVE, and IMO have experienced immense variation between the years 2000 and 2024.
- The maximum stock price was registered by IMO (Imperial Oil) by breaching over 100 in the year 2024, while the minimum was recorded by CVE (Cenovus Energy).

- There is a steep dip in prices around 2020 owing to the potential COVID-19 oil price crash.
- Each of the stocks has regained itself since the year 2021, with IMO recording the greatest growth.

Impact of GDP on Stock prices.

(Refer to `GDP_query` in our submitted notebook)

Result:



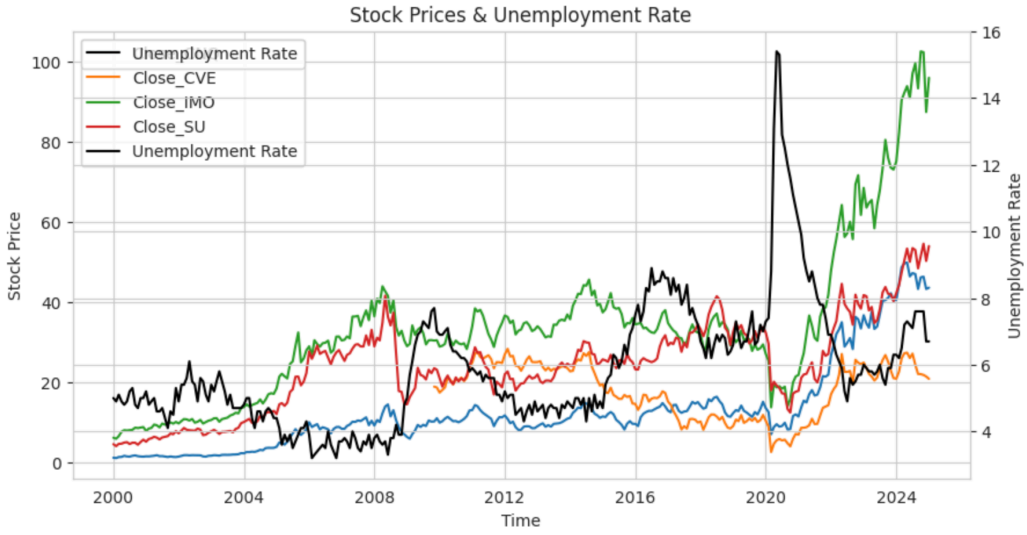
Findings:

- The stock prices of CNQ, SU, IMO, and CVE all have a long-term upward trend, with IMO recording the highest average price.
- The sharp drop in GDP around 2020 is the COVID-19 crisis corresponding to the lowest point before strongly recovering
- Both the stock prices and GDP have recovered strongly since 2021, with GDP surpassing the pre-2020 point.
- CVE has the lowest average share price, whereas SU and CNQ record consistent but slower growth compared to IMO.

Impact of Unemployment on Stock Prices

(Refer to `unemployment_query` in our submitted notebook)

Result:



Findings:

- Unemployment rose sharply in the vicinity of 2020, with the peak being over 15%, which coincided with the same period of all stock prices dropping.
- IMO stock (green line) was the highest since 2021, crossing 90, while CVE was the lowest in general.
- Stock prices bounced back strongly post-2020 as unemployment dropped, which showed reverse correlation.
- SU and CNQ showed consistent rises, with SU hitting about 50 by 2024, while unemployment remained constant at 7%.

Which stock has the highest growth?

Calculated the year-over-year (YoY) growth of stock prices for CNQ, CVE, SU, and IMO, ranked them by total growth. *(Refer to* `yoy_growth_query` *in our submitted notebook)*

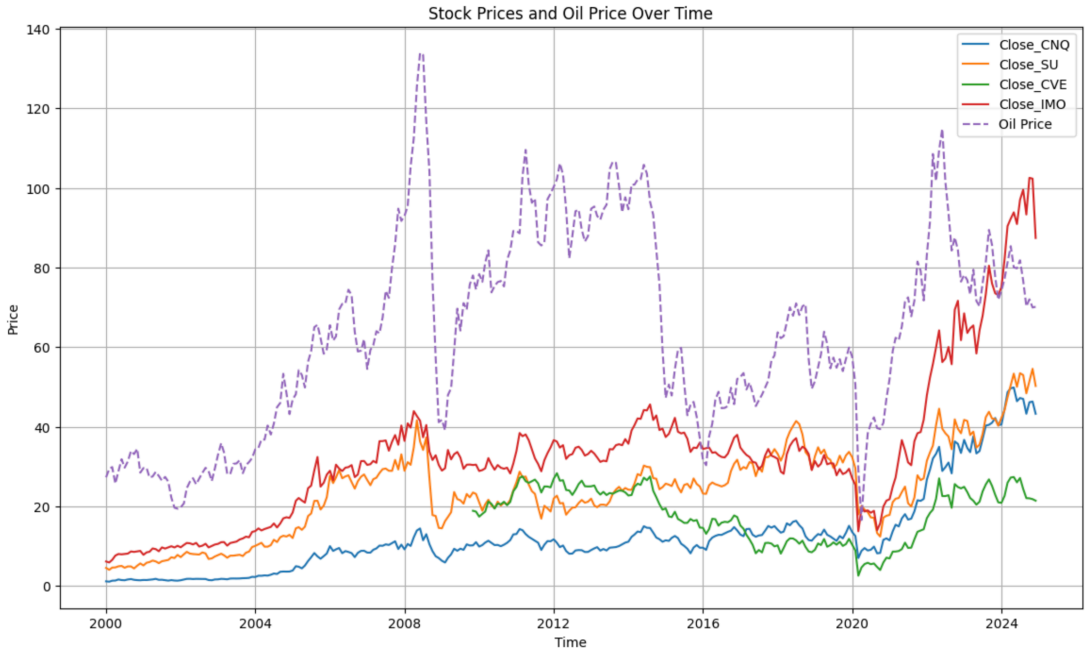
Result:

	stock_name	total_growth	overall_rank
0	CNQ	3162.687377	1
1	IMO	1178.521847	2
2	SU	1048.991934	3
3	CVE	338.206379	4

Impact of oil price on stock prices

(Refer to `oil_price_query` in our submitted notebook)

Result:



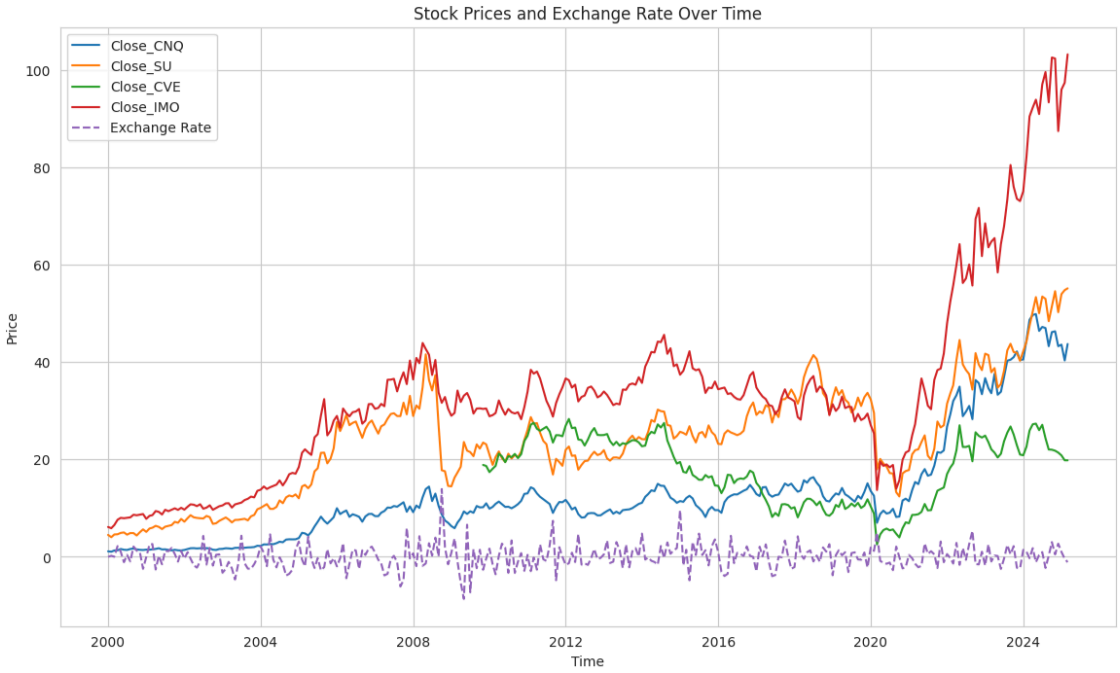
Findings:

- Oil prices (dashed purple) peaked above 130 and dropped near 20 at the lowest.
- IMO stock (red) had the highest stock price, while CNQ (blue) had the lowest overall.
- A significant dip occurred around 2020, followed by a strong recovery in prices.

Exchange rate and Stock Prices

(Refer to `exchange_query` in our submitted notebook)

Result:



Findings:

- IMO (red) had the highest stock price, while CNQ (blue) had the lowest
- A sharp dip occurred around 2020, followed by a strong upward trend.
- The exchange rate remained relatively stable with minor fluctuations.

Impact of Inflation on stock Prices

(Refer to `inflation_query` n our submitted notebook)

Result:

Findings:

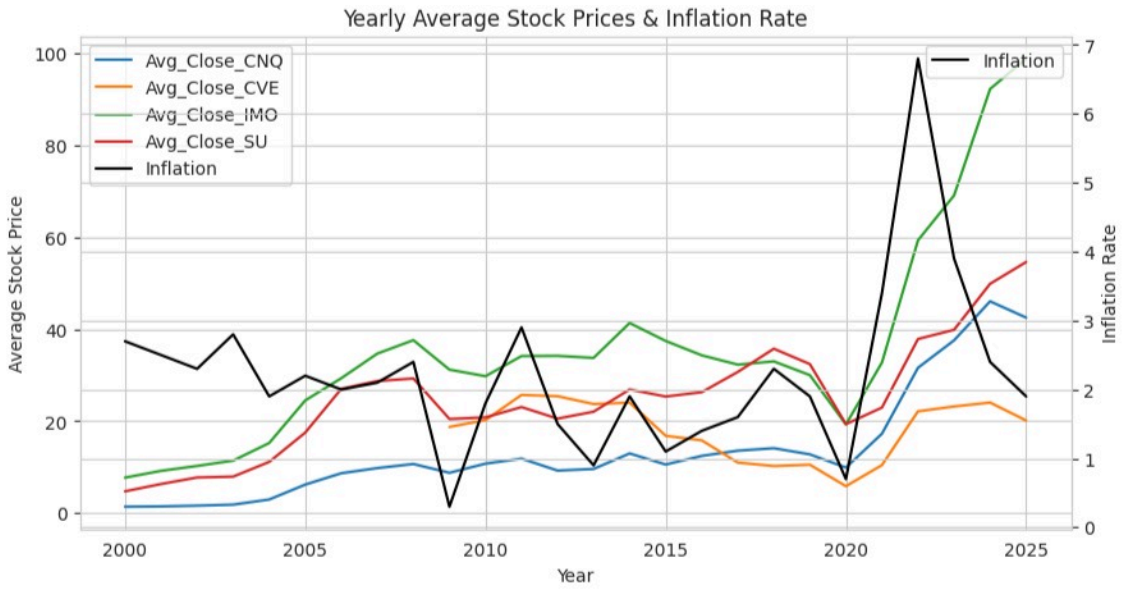
- Stock prices and inflation dipped in 2020 but surged afterward, showing a strong recovery.
- Inflation spiked dramatically post-2020, aligning with a sharp rise in stock prices, especially for IMO and SU
- Inflation declined after its peak, but stock prices remained elevated, suggesting resilience in the oil sector.

Yearwise impact of Inflation on Stock prices

(Refer to `inflation_query` in our submitted notebook)

Result:

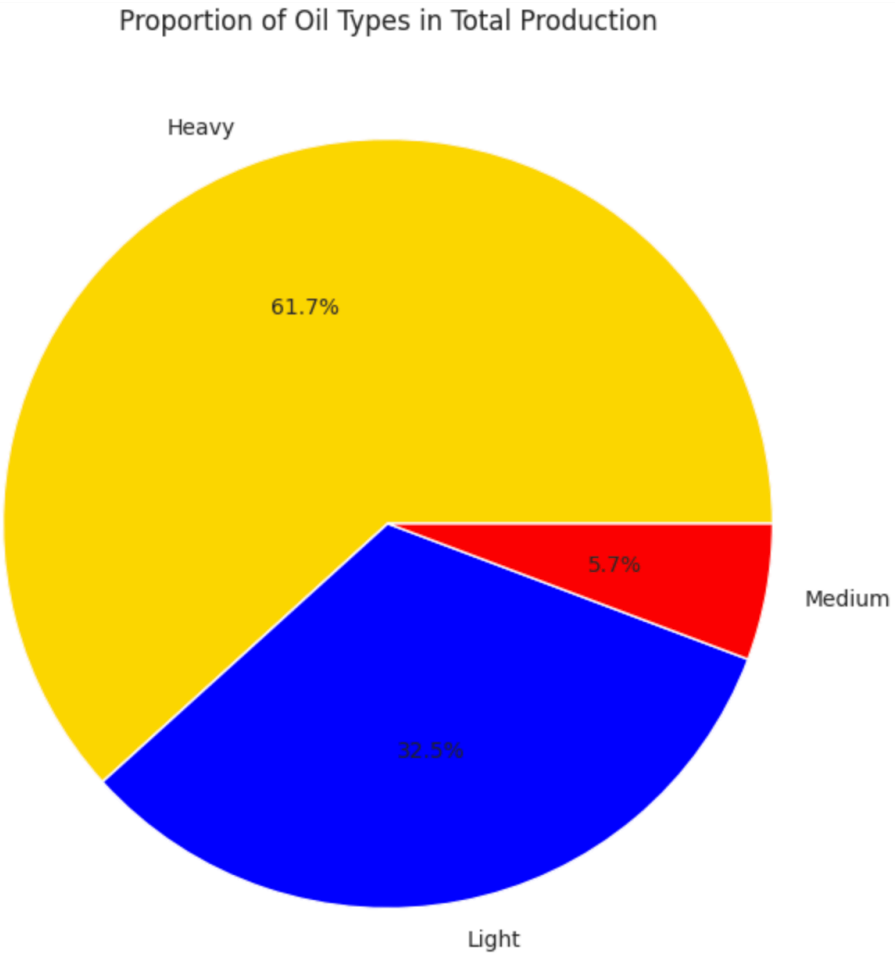
	Year	Avg_Close_CNQ	Avg_Close_CVE	Avg_Close_IMO	Avg_Close_SU	Canada
0	2022	31.611325	22.145866	59.329158	37.868591	6.8
1	2023	37.571489	23.201345	69.073570	39.872421	3.9
2	2021	17.279444	10.455362	32.754750	22.966569	3.4
3	2011	11.863678	25.711076	34.160012	23.070944	2.9
4	2003	1.859604	NaN	11.429150	7.948847	2.8



Which oil types dominate exports in total oil production?

(Refer to `oil_type_query` in our submitted notebook)

Result:



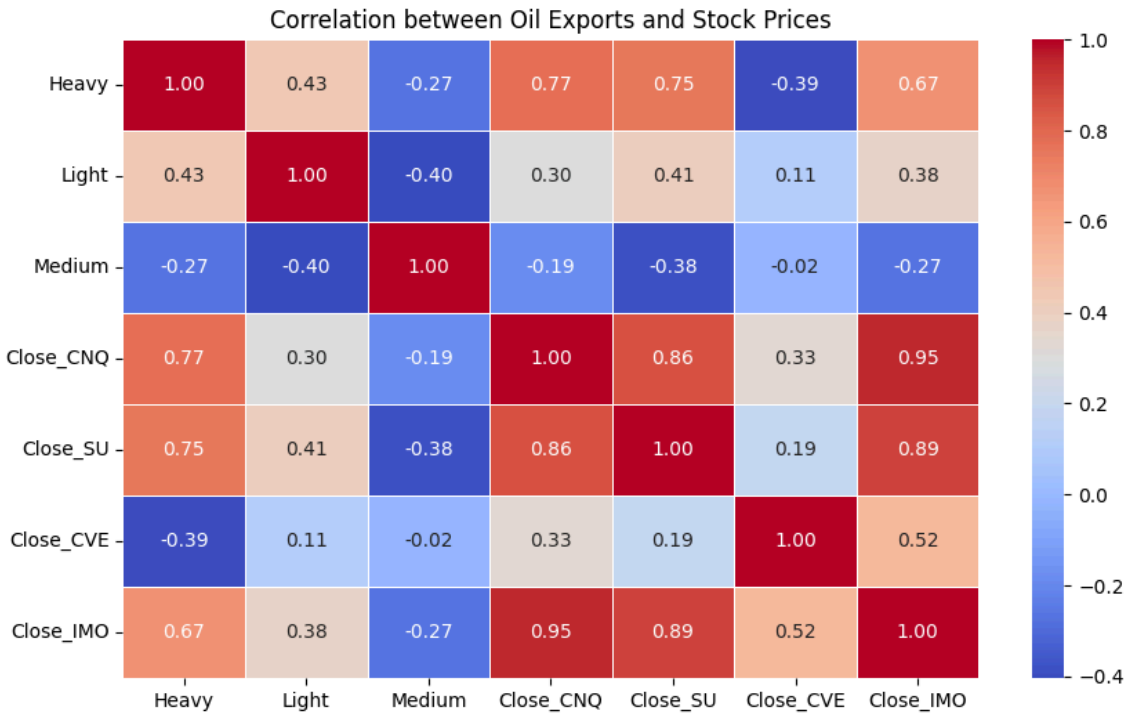
Findings:

- Heavy oil dominates production, meaning that the industry is primarily extracting and processing this type of oil.
- Light oil still plays a significant role, though much smaller than heavy oil.
- Medium oil is the least produced, suggesting either limited availability or lower demand.

How stock prices of oil companies relate to oil exports over time?

(Refer to the `oil_type_correlation_query` in our submitted notebook)

Result:



Findings:

- Close_CNQ and Close_IMO have the highest correlation (0.95), indicating a strong relationship between their stock prices.
- Medium oil exports and Close_CVE have the lowest correlation (-0.02), meaning that Medium oil exports have little to no impact on CVE's stock price
- Light oil exports have moderate correlations with stock prices, with the highest being Close_SU (0.41). This suggests Light oil has a weaker impact on stock performance compared to Heavy oil.
- Medium oil exports have negative correlations with all stock prices, especially Close_SU (-0.38) and Close_CNQ (-0.19), indicating that higher Medium oil exports might be linked to lower stock prices for these companies.

Steps to combine dataset

-Stock Data (Closing prices, volume, daily returns) was stored in a table (stock_merged).

-Oil Production Data was stored in production.

-Oil Price Data was store in prices

-Macroeconomic Indicators (GDP, inflation, Unemployment Rate) were stored in individual tables. -Geopolitical Events were manually annotated for specific dates in a separate reference table.

-Data Cleaning & Standardization:

-Ensured all tables used a common date format (YYYY-MM-DD).

-Removed missing values and performed data imputation where necessary.

-Normalized stock prices and macroeconomic indicators to enable meaningful comparisons.

Our Learnings and Discussion:

While acquiring the data and importing it into an SQL database, we learned working with various formats of data such as CSV, JSON, and Excel while keeping data validation intact and consistent. Error handling mechanisms were implemented to identify inconsistencies and preserve data integrity. Overall, this experience provided us with hands-on experience in properly organizing and managing real-world datasets for analysis.

Here is the detailed output of our individual learnings:

Romith- Through this project, I gained valuable experience in integrating diverse datasets—macroeconomic indicators, oil export volumes, and company stock prices—to uncover meaningful correlations and insights. I enhanced my skills in SQL for data extraction and Python for time series analysis, data visualization, and statistical calculations like volatility and growth. This project highlighted the importance of aligning temporal data and dealing with real-world data inconsistencies. Looking back, incorporating tools like Power BI or advanced techniques such as ARIMA forecasting could improve the depth of analysis. In the future, I would extend this project by exploring sentiment analysis and global oil price trends to enrich the findings.

Hawa - I learnt how to use inner join in a query to join multiple datasets and extract information needed to answer specific questions. For further studies, i would like to analyze if there is a long term relationship between stock prices, inflation and exports using Vector Error Correction Model. I would like to answer whether inflation and exports significantly drive stock prices in the long run.

Kajal I now have a better understanding of a variety of data analysis aspects from the acquisition of data and optimization of SQL. I realized the importance of handling large datasets efficiently, making use of SQL for accelerating computation, and generating improved financial trend visualizations. Looking back, cloud databases such as Google BigQuery or AWS RDS would have added scalability and performance. There would have been automated ETL (Extract, Transform, Load) pipelines included in data processing to make it easier. I would focus on streamlining our data cleaning procedures and exploring alternative statistical models for better predictive accuracy on redoing the project. In the coming years, one exciting addition would be the introduction of machine learning techniques for prediction of stock price, sentiment from financial news, and expanding the dataset to incorporate global markets so that economic influence on stock dynamics is analyzed completely.

Ali- For future work, incorporating machine learning techniques to predict future stock prices based on historical data and identified trends would be a valuable exploration. Furthermore, understanding the interplay between macroeconomic indicators and stock prices can also be extended by analyzing sector-specific impacts and how companies in different industries react to changes in the stock market. Overall, this multifaceted analysis sets the stage for more complex predictive modeling and market behavior studies.

Conclusion

This project offered a thorough exploration of stock market behavior by combining financial data with macroeconomic indicators and geopolitical events. The analysis underscored how external factors significantly influence market dynamics, particularly highlighting the heightened volatility during geopolitical crises and the observable correlations between economic indicators and stock performance. Looking ahead, expanding the dataset to include additional sectors or international markets, as well as applying advanced forecasting models, could further enrich the analysis and deepen our understanding of global financial trends.

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